

AEROSPACE ENGINEERING · GEORGIA TECH

Dean Cardner

Engineering Portfolio

Valves, fluid systems, and test engineering for liquid rocket propulsion. Three successive responsible engineer roles on a record-breaking KeroLOX student team, as well as test operations internships at NASA JSC and MSFC.

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B.S. AEROSPACE ENGINEERING, MAY 2028 · 4.0 GPA

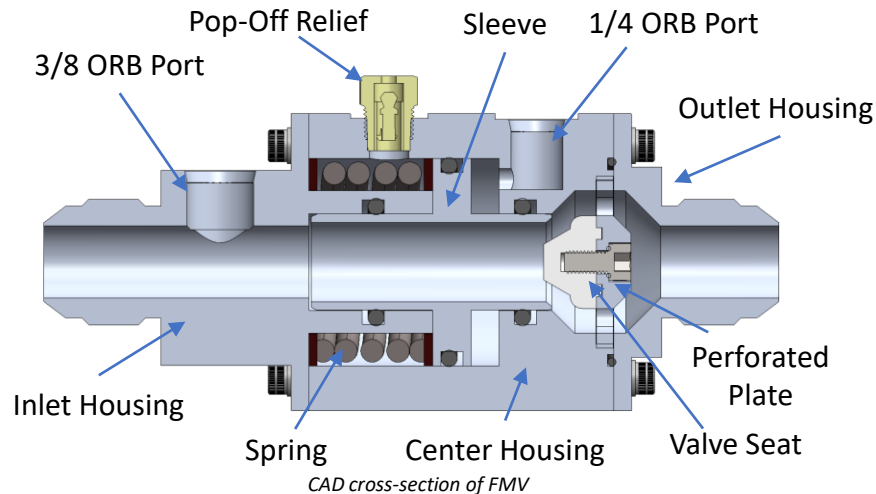


Experience

AUG 2026 –	Fuel Main Valve Responsible Engineer Yellow Jacket Space Program Own development and testing of a custom kerosene sleeve valve.	01
MAY – AUG 2026	Valve Test Intern NASA Marshall Space Flight Center (Amentum) Valves, Actuators, Lines & Ducts Branch. Component and thruster testing.	04
JAN – MAY 2026	Test Operations Intern NASA Johnson Space Center (OSTEM) Systems Test Branch. Thermal-vacuum testing of spaceflight hardware.	04
AUG 2025 – JAN 2026	Launch Feed System Responsible Engineer Yellow Jacket Space Program Owned a five-subsystem ground fluids system through the launch campaign.	02
JAN – AUG 2025	LOX Valve Actuator Responsible Engineer Yellow Jacket Space Program Designed, machined, and tested a spring-return fail-safe actuator.	03

01 Fuel Main Valve (FMV) - Overview

RESPONSIBLE ENGINEER · OVERVIEW · PART 1 OF 2



The hardware

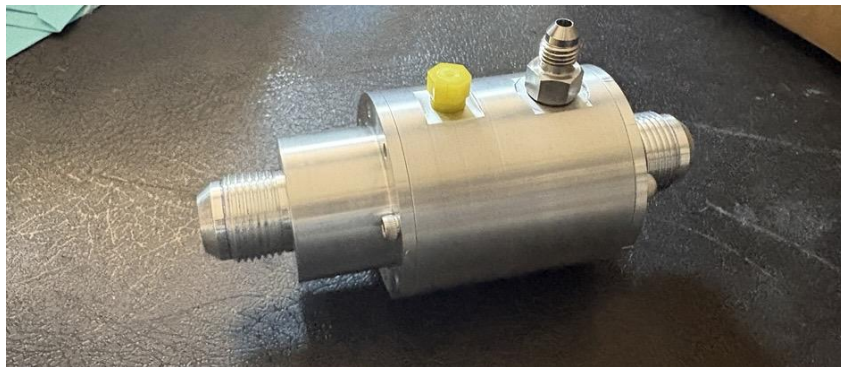
A custom kerosene sleeve valve serving as the fuel main valve on our spaceshot KeroLOX vehicle. Prominent features include Aluminum 7075 valve housings, a PEEK valve seat, ports for the flow of kerosene and nitrogen gas pneumatics, and a compression spring.

MY SCOPE

I inherited the design and the valve design trades. I own manufacture of the qualification article, the qualification campaign, and any design changes required before I manufacture and test a flight article.

Analysis support

Set up a STAR-CCM+ CFD simulation to characterize pressure drop across the valve at the design condition of 560 psig inlet and 1.78 kg/s kerosene mass flow. The simulation found a predicted pressure drop of 15.0 psid, which will be verified in testing.



FMV valve housing

Fuel Main Valve (FMV) - Manufacturing and Qualification

Test article

Parts machined from Aluminum 7075 bar stock on manual mill and lathe at Georgia Tech's AE Machine Shop.

Procedures

Drafted the procedures covering the full sequence:

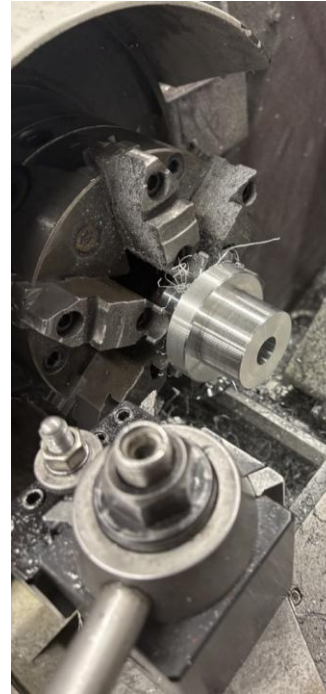
- Proof pressure testing
- Leak testing
- Flow testing at design conditions
- Lifecycle testing

These tests will be supported by YJSP's Component Test Stand later this fall.

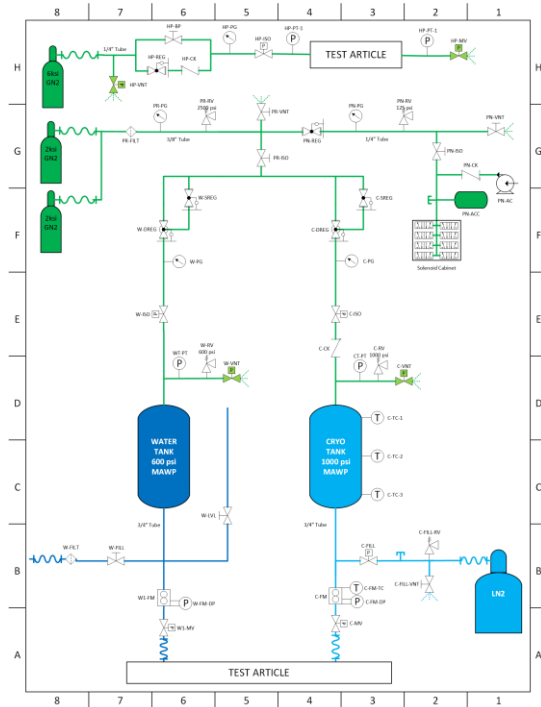
After the conclusion of qualification testing, I will own any design changes identified before manufacturing and acceptance testing the flight article.

STATUS

› I am currently finalizing the qualification article, including the manufacturing of the sleeve and perforated plate. The parts are to be cleaned, then receive an Alodine 1201 coating to increase corrosion resistance. Then, I will hold a Test Readiness Review, and qualification testing can begin.



Sleeve manufacturing on a manual lathe



Component Test Stand P&ID. The Fuel Main Valve will be tested using this system.

02 Launch Feed System — Fluids GSE for a KeroLOX Rocket

RESPONSIBLE ENGINEER · SYSTEM OWNERSHIP · PART 1 OF 3



The Launch Feed System during test operations



Vespula static fire

ROLE	Launch Feed System RE
TEAM	Yellow Jacket Space Program, <i>Vespula</i> Vehicle Program
DATES	Aug 2025 – Jan 2026

The system

Five subsystems and over fifty valves handling liquid oxygen, kerosene, 6000 psi nitrogen, pneumatics, and CO2 fire suppression. Supported the static fire and the first launch attempt of *Vespula*, a record-breaking KeroLOX vehicle that reached an altitude of 56,590 feet.

What I owned

- Testing and operation of the full system
- Ground-side pad setup, valve actuation and leak checks
- Operation of the pressure-reducing panel
- Avionics harnessing integration
- LOX handling as designated cryo-warrior

Operating the system

Led seven propellant and pressurant fill operations across the campaign, with no system failures or safety incidents. Supported a successful static fire and completed full verification of the ground fluids system ahead of launch.

WHAT OPERATING IT INVOLVED

- Operated the pressure-reducing panel, verifying K-bottle pressures and regulating 6000 psi nitrogen for use as pressurant
- Handled liquid oxygen in cryogenic PPE as designated cryo-warrior
- Safed the system after each test by actuating manual and remote vents to clear trapped gas and cryogenic liquid



Procedure execution during cryogenic testing



Directing remote valve actuations from the pad

System upgrades

50%

CUT FROM PANEL PRIMING TIME

HIGH-PRESSURE VENT LINE

Designed a non-propulsive vent line for the 6000 psi K-bottle manifold, then cut, bent, and flared the stainless tube stock to fabricate it. This vent line streamlined the process of checking K-bottle pressures and priming the system for pressurant fill, reducing setup time by 50%.

KEROSENE FUEL FILL SYSTEM

Redesigned and plumbed the fuel fill system, selecting a particulate filter and an 18 GPM pump.



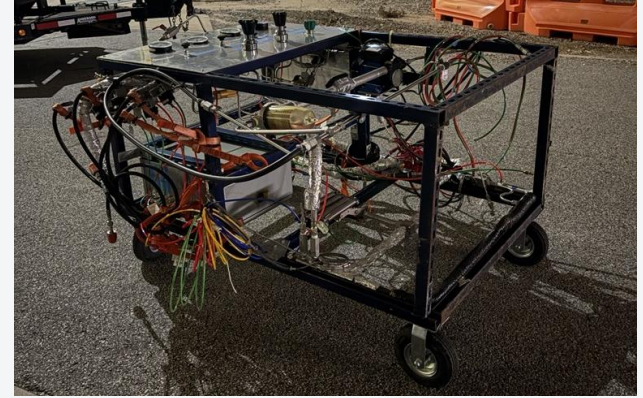
High-pressure GN2 vent line

P&ID REVISION

Revamped reference designators and P&IDs across every subsystem.

MOBILITY

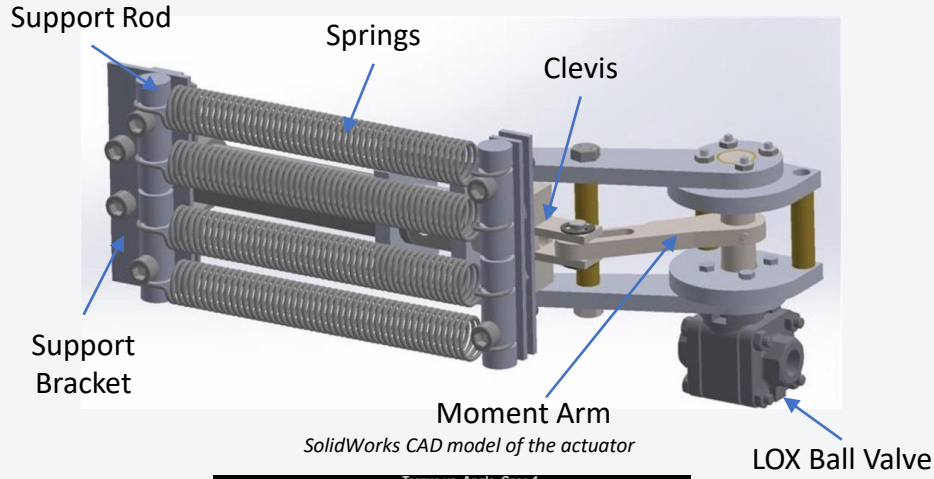
Selected and installed caster wheels, cutting pad setup time.



System with integrated caster wheels

03 LOX Valve Spring-Return Actuator

RESPONSIBLE ENGINEER · DESIGN & SIZING · PART 1 OF 2

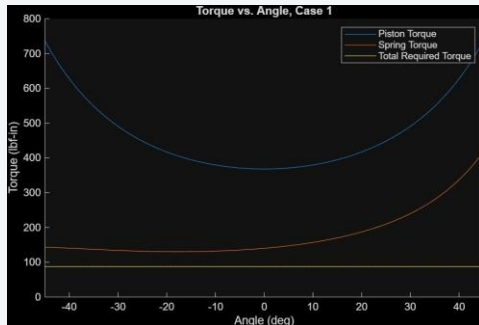


The problem

The LOX ground fill valve had to fail closed. Losing pneumatic supply with the valve open risks excess propellant loss on a live pad, so the fail-safe could not itself depend on power or gas.

Sizing

Designed a scotch-yoke pneumatic actuator, which included extension springs that would close the valve in case of loss of pneumatics. Ran a force and torque analysis from first principles, accounting for pneumatics pressure, piston bore size, and moment arm geometry. Modeled six custom components in SolidWorks.



Torque vs. Angle graph for the scotch-yoke actuator

Manufacture & test

Manufactured in-house

Machined all six custom components — clevis, support rods, brackets — on manual mill, lathe, and waterjet.

Test director

Wrote and ran the LOX subsystem procedures: nominal valve actuation under cryogenic conditions, and leak checking the three cryogenic valves and the oxidizer quick disconnect when cold.

- Setup: harness integration, solenoid click checks, actuation checks
- Execution: leak checks, initial flow, long-duration flow



Successful actuation of the final assembly



Cryogenic testing with a liquid nitrogen dewar

VERIFIED

Opens and closes under cryogenic conditions

VERIFIED

Returns to safe state on loss of pneumatics

VERIFIED

Zero actuator failures through the campaign

04 NASA Test Operations Internships

TWO TERMS · JAN - AUG 2026

MARSHALL SPACE FLIGHT CENTER

Valve Test Intern

Amentum · Valves, Actuators, Lines & Ducts · May – Aug 2026

- Supported component and thruster testing in the Component Development Area, NASA's valve test lab
- Selected components, drafted a test schematic, ran pressure drop calculations, and sized relief valves for a hydrogen-oxygen gas thruster
- Designed and fabricated a 0–200 lb-in strain gauge torque transducer: gauge installation, stress calculations, soldering, and LabVIEW data acquisition
- Designed and 3D printed a Creo mold for casting a custom ceramic nozzle

JOHNSON SPACE CENTER

Test Operations Intern

OSTEM · Systems Test Branch · Jan – May 2026

- Supported thermal-vacuum testing of spaceflight and lunar surface hardware
- Designed the pressurization and leak check system for a human-rated spacesuit weight-relief device: P&ID, concept of operations, regulator selection, and relief valve sizing
- Fabricated the pressurization cart, assembling an 80/20 frame and mounting the tank and valves

Skills & tools

FLUID SYSTEMS & CRYOGENICS

- LOX, RP-1, and 6000 psi GN2 systems
- Regulator and relief valve sizing, pressure drop calcs
- P&ID authoring, reference designators, ConOps
- Cryogenic handling, leak checks, system safing

TEST ENGINEERING

- Procedure authoring and test execution
- Proof, leak, flow, and lifecycle qualification testing
- Strain gauges, torque transducers, LabVIEW DAQ
- Thermal-vacuum and cryogenic test support

DESIGN & ANALYSIS

- SolidWorks, Creo — assemblies, drawings, design reviews
- STAR-CCM+ CFD; MATLAB and Python; CoolProp; Visio
- Force and torque analysis from first principles
- Component selection and sizing

MANUFACTURING

- Manual mill, lathe, waterjet — Al 7075-T6 and stainless
- Tube fabrication: cutting, bending, flaring, facing
- 3D printing, soldering, 80/20 frame assembly

THANKS FOR READING

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